

RESCHE BALAGANESH

AI Engineer | LLM Applications | Multi-Agent Systems | AWS

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EDUCATION

Western University

Bachelor of Engineering, Software Engineering

September 2020 – April 2025

PROFESSIONAL EXPERIENCE

Bank of Montreal

Toronto, ON, Canada

Software Developer | Assigned Tech Lead

June 2025 – Present

- Architected a stateful **multi-agent QA automation system** using **LangGraph** and **AWS Bedrock** that eliminated manual test authoring entirely across the application codebase
- Designed Agent 1 with **S3 directory and file read tools** to identify relevant files, and scoped Agent 2 to a **single file read tool**, enforcing least-privilege **tool access** and minimizing context consumption in test generation
- Selected as **technical lead** for a critical enterprise-wide initiative, owning environment stability across **30+ distributed systems** by triaging failures at the **infrastructure, service, and integration layer** ahead of production releases
- Drove **cross-team incident resolution** by analyzing system dependencies, coordinating across **3–5 engineering teams** simultaneously, and tracking outstanding issues in parallel to prevent environment blockers from escalating into production delays

Royal Bank of Canada

Toronto, ON, Canada

Patent-Pending AI Evaluation Platform - Capstone

September 2024 – April 2025

- Architected a **serverless AWS** investment evaluation platform using **API Gateway, Lambda, SQS, DynamoDB, S3**, and **AWS CDK**, enabling parallel AI-driven scoring across thousands of census tracts at scale
- Built an **LLM-powered decision engine** using **OpenAI** that converts natural-language user input into structured weighted scoring factors, distributing evaluation work across SQS for independent parallel assessment of each census tract
- Engineered **context-aware scoring logic** that feeds existing scores into each evaluation as a calibration reference, ensuring the model scores regions on a consistent scale and preventing uncalibrated outputs across **1,000+ investment opportunities** per pipeline run
- Delivered a **conversational AI assistant** allowing users to collaboratively shape how the scoring engine interprets and weighs investment criteria before execution, creating a **human-in-the-loop** configuration layer, alongside a **React heatmap** that cut stakeholder analysis time by **96%**
- System architecture selected by RBC to pursue a **patent** for the investment evaluation workflow and platform design

Bank of Montreal

Toronto, ON, Canada

Junior Cloud Analyst, DevOps Automation Team

May 2023 – August 2024

- Engineered and maintained custom **Infrastructure as Code** modules using **TypeScript, Terraform**, and **AWS CDK**, enforcing security and operational compliance standards while delivering **USD \$275,000** in annual cost savings
- Designed and maintained **GitHub Actions CI/CD** workflows for both **AWS** and **Azure** environments using AWS CDK and CDKTF, supporting cloud infrastructure deployments for over **20,000 developers** across the organization
- Led the enterprise **Artifactory SaaS Migration**, transitioning workflows from on-premises Artifactory to Artifactory SaaS and architecting workflow designs with cross-functional teams to implement automated security and billing compliance checks, improving operational efficiency by **50%** with no disruption to developer workflows
- Delivered a new major version of BMO's bank-wide shared **laC modules**, resolving cross-version compatibility issues, fixing defects, and completing unit and integration testing across **80+ cloud resource modules** prior to rollout

AI PROJECTS

Hive | Autonomous Incident Diagnosis Platform

2026 – Present

- Designed a **LangGraph ReAct agent** on **OpenAI** that autonomously traverses a live **call graph**, deciding which services to investigate based on anomaly signals and driving **per-service diagnosis** to pinpoint root causes autonomously
- Equipped the agent with **diagnostic functions** for trace retrieval, topology queries, and input/output inspection, enabling it to gather and reason over distributed failure evidence and synthesize findings into a plain-English **incident report**
- Built a **low-config Python SDK** that propagates **trace IDs** across services and captures **internal call graphs** via runtime instrumentation, supplying the agent with real-time investigation context at zero latency cost
- Architected a serverless **AWS control plane (API Gateway, SQS, Lambda, DynamoDB)** that ingests distributed events and maintains a continuously updated **service topology** consumed by the agent at diagnosis time

Smart Trader | AI-Assisted Stock Screening Platform

2026 – Present

- Built a production **RAG pipeline** on **AWS ECS** using **LangChain** and **Pinecone**, generating grounded BUY, WAIT, and PASS recommendations derived from real-time technical signals and market news across **10,000+ stocks**
- Engineered a **dual-retrieval architecture** spanning user-specific preference memory and a **shared news knowledge base (Pinecone & DynamoDB)**, improving recommendation relevance and eliminating ungrounded model responses
- Developed an **LLM-driven summarization layer** to produce structured, retrieval-optimized document embeddings, reducing vector index footprint and token costs while retaining all decision-critical context

- Architected a tiered data store using **DynamoDB** and **S3** as the system of record for full signal and news payloads, with **Pinecone** indexing lightweight summaries and metadata pointers for efficient retrieval

SKILLS & CERTIFICATIONS

Certifications: AWS Developer Associate, AWS Solutions Architect Associate

Cloud & DevOps: AWS, Lambda, DynamoDB, APIGW, SQS, ECS, Azure, Terraform, AWS CDK, GitHub Actions, CI/CD, Docker, Kubernetes, Dynatrace, JIRA, Agile

AI / LLM Engineering: LangChain, LangGraph, LangSmith, AWS Bedrock, RAG, Pinecone, Embeddings, Semantic Similarity Search, OpenAI API, Multi-Agent Systems, Prompt Engineering

Languages: Python, TypeScript, JavaScript, Java, SQL, Bash